

The Symmetric Group and Cayley's Theorem

Road map

This chapter introduces the symmetric group of a finite set and carefully distinguishes it from the symmetry group of a geometric object. It then turns the left translations of an arbitrary finite group into a permutation group. The resulting faithful action proves Cayley's theorem: every finite group of order n is isomorphic to a subgroup of S_n .

3.1 The symmetric group of a finite set

Definition 3.1: Symmetric group

Let X be a finite set. The *symmetric group on X* is

$$S_X = \{f : X \longrightarrow X : f \text{ is bijective}\}.$$

Its operation is composition:

$$(f \circ g)(x) = f(g(x)).$$

Thus g acts first and f acts second.

Let us verify the group axioms, including the facts that are often left implicit.

- If $f, g : X \rightarrow X$ are bijections, then $f \circ g$ is a bijection: its inverse is $g^{-1} \circ f^{-1}$. Hence S_X is closed.
- Composition of maps is associative:

$$((f \circ g) \circ h)(x) = f(g(h(x))) = (f \circ (g \circ h))(x)$$

for every $x \in X$.

- The identity map $\text{id}_X(x) = x$ is the identity element.
- Every $f \in S_X$ has its inverse bijection $f^{-1} \in S_X$.

Consequently S_X is indeed a group.

Symmetric group versus symmetry group

The terms are related but are not synonymous.

- The *symmetric group* S_X contains all permutations of the set X .
- A *symmetry group* contains transformations that leave a chosen geometric or other structured object invariant.

For example, the symmetry group of a regular polygon is a dihedral group, whereas S_X forgets geometry and permits every bijection of its underlying set.

If X has n elements, choose a labelling

$$X = \{x_1, \dots, x_n\}$$

and identify x_i with i . Relabelling gives an isomorphism

$$S_X \cong S_n,$$

where

$$S_n := S_{\{1, 2, \dots, n\}}.$$

The group S_n is also called the *permutation group on n letters*. A permutation may be written as a bijection

$$f : \{1, \dots, n\} \longrightarrow \{1, \dots, n\}.$$

Proposition 3.1: Order of S_n

$$|S_n| = n!.$$

证明. There are n choices for $f(1)$. Once that value is chosen, bijectivity leaves $n - 1$ choices for $f(2)$, then $n - 2$ choices for $f(3)$, and so on, ending with one choice for $f(n)$. By the multiplication principle,

$$|S_n| = n(n - 1)(n - 2) \cdots 2 \cdot 1 = n!.$$

■

Example 3.1: S_3 and the triangle group

S_3 has $3! = 6$ elements. It has the same multiplication structure as the symmetry group D_3 of an equilateral triangle:

$$S_3 \cong D_3.$$

Indeed, every permutation of the three vertices is realized by exactly one rigid symmetry of the triangle. Under this correspondence, the 3-cycles are rotations and the transpositions are reflections. Hence their Cayley tables agree after the elements are relabelled.

3.2 Left translations as permutations

Let

$$G = \{a_1, \dots, a_n\}, \quad |G| = n,$$

be a finite group. For each $a \in G$, recall the left translation

$$\ell_a : G \longrightarrow G, \quad \ell_a(x) = ax.$$

The cancellation argument from Chapter 2 shows that every ℓ_a is bijective. Thus each left translation is an element of the symmetric group S_G .

Definition 3.2: The left regular permutation group

Define

$$L_G := \{\ell_a : a \in G\}.$$

This is the set of all left translations of G .

Proposition 3.2

L_G is a group under composition.

证明. We check every group axiom.

Closure. Let $a, b \in G$. For every $x \in G$,

$$\begin{aligned} (\ell_a \circ \ell_b)(x) &= \ell_a(\ell_b(x)) \\ &= \ell_a(bx) \\ &= a(bx) \\ &= (ab)x \\ &= \ell_{ab}(x). \end{aligned}$$

Two maps are equal when they have the same value at every input, so

$$\boxed{\ell_a \circ \ell_b = \ell_{ab}}.$$

Since $ab \in G$, the composite belongs to L_G .

Identity. For every $x \in G$,

$$\ell_e(x) = ex = x.$$

Hence $\ell_e = \text{id}_G$, and

$$\ell_e \circ \ell_a = \ell_a = \ell_a \circ \ell_e.$$

Inverses. Using the composition rule,

$$\ell_{a^{-1}} \circ \ell_a = \ell_{a^{-1}a} = \ell_e, \quad \ell_a \circ \ell_{a^{-1}} = \ell_{aa^{-1}} = \ell_e.$$

Therefore

$$(\ell_a)^{-1} = \ell_{a^{-1}}.$$

Associativity. Composition of functions is associative:

$$\ell_a \circ (\ell_b \circ \ell_c) = (\ell_a \circ \ell_b) \circ \ell_c.$$

Thus all group axioms hold. ■

3.3 The group G and the group L_G

Consider the map

$$\Phi : G \longrightarrow L_G, \quad a \longmapsto \ell_a.$$

Proposition 3.3

Φ is a bijection and preserves multiplication:

$$\Phi(ab) = \Phi(a) \circ \Phi(b).$$

证明. *Preservation of multiplication.* The composition calculation above gives

$$\Phi(a) \circ \Phi(b) = \ell_a \circ \ell_b = \ell_{ab} = \Phi(ab).$$

Surjectivity. By the definition of L_G , every one of its elements has the form $\ell_a = \Phi(a)$ for some $a \in G$. Thus Φ is onto.

Injectivity. Suppose $\Phi(a) = \Phi(b)$, so $\ell_a = \ell_b$. Equal maps agree at e , and therefore

$$a = \ell_a(e) = \ell_b(e) = b.$$

Equivalently, equality at every $x \in G$ gives $ax = bx$, and right cancellation gives $a = b$. Thus Φ is one-to-one. ■

Definition 3.3: Group isomorphism

A map $\varphi : G \rightarrow H$ is a *group isomorphism* if it is bijective and

$$\varphi(ab) = \varphi(a)\varphi(b) \quad \text{for all } a, b \in G.$$

If such a map exists, we write $G \cong H$.

The proposition proves

$$G \cong L_G.$$

It also explains in a literal way why the two groups have the same Cayley table structure. The element a corresponds to ℓ_a , and

$$ab = c \iff \ell_a \circ \ell_b = \ell_c.$$

Thus every entry of the Cayley table of G becomes the corresponding entry of the Cayley table of L_G .

Since Φ is bijective, there is a one-to-one correspondence

$$\begin{array}{ccc} G & \longleftrightarrow & L_G \\ a & \longleftrightarrow & \ell_a \end{array}$$

and consequently

$$|G| = |L_G| = n.$$

Example 3.2: The regular copy of $\mathbb{Z}_3$ inside S_3

Let $G = \{e, a, a^2\}$ with $a^3 = e$. In cycle notation for permutations of the set $\{e, a, a^2\}$,

$$\ell_e = \text{id}, \quad \ell_a = (e \ a \ a^2), \quad \ell_{a^2} = (e \ a^2 \ a).$$

Hence

$$L_G = \{\text{id}, (e \ a \ a^2), (e \ a^2 \ a)\}.$$

It is a three-element subgroup of the six-element group $S_G \cong S_3$, and it is isomorphic to $\mathbb{Z}_3$.

3.4 Cayley's theorem

Because every left translation is a permutation of the underlying set G , we have

$$L_G \subseteq S_G.$$

The group operations agree: both are composition of maps. Therefore the subgroup test, or the group verification above, gives

$$L_G \leq S_G.$$

If $|G| = n$, then a labelling of the elements of G gives

$$S_G \cong S_n.$$

Theorem 3.1: Cayley's theorem

Every finite group G of order n is isomorphic to a subgroup of S_n . More precisely,

$$G \cong L_G \leq S_G \cong S_n.$$

证明. The map

$$\Phi : G \longrightarrow S_G, \quad a \longmapsto \ell_a,$$

preserves multiplication because $\ell_a \circ \ell_b = \ell_{ab}$. It is injective because $\ell_a = \ell_b$ implies

$$a = \ell_a(e) = \ell_b(e) = b.$$

Its image is exactly L_G , which is a subgroup of S_G . Thus Φ is an isomorphism from G onto L_G . Finally, after labelling the n elements of G by $1, \dots, n$, the group S_G becomes S_n . ■

The theorem also records the size comparison

$$|G| = |L_G| = n, \quad |S_G| = n!.$$

When $G \neq \{e\}$, the subgroup L_G is nontrivial. For $n \geq 3$, most permutations of the underlying set are not left translations: L_G has only n elements, while S_G has $n!$ elements.

The theorem beyond finite groups

The same proof works for an arbitrary group G , finite or infinite:

$$G \cong L_G \leq S_G,$$

where S_G now means the group of all bijections $G \rightarrow G$. No counting by $n!$ is then intended. The handwritten note phrases this as the infinite, discrete version of Cayley's theorem; algebraically, no topology is needed for the proof.

What to retain from this chapter

1. S_X is the group of all bijections $X \rightarrow X$, with composition. If $|X| = n$, then $S_X \cong S_n$ and $|S_n| = n!$.
2. Left translations satisfy $\ell_a \circ \ell_b = \ell_{ab}$, $\ell_e = \text{id}_G$, and $(\ell_a)^{-1} = \ell_{a^{-1}}$.
3. The correspondence $a \leftrightarrow \ell_a$ is bijective and preserves multiplication, so $G \cong L_G$.
4. Since $L_G \leq S_G \cong S_n$, every group of order n occurs as a subgroup of S_n .